How many clusters? A 5-state vs. 7-state comparison of the VAE–HMM convection model

Complementary report

May 22, 2026

1 Motivation

The paper analyses convective organization through a VAE–HMM with $K = 7$ latent states. A natural question is whether the headline results—(i) a precipitation/organization plane of variability, (ii) an organized, high-PW cluster with deep coherent mass flux, (iii) a two-regime leading non-stationary mode v_2, and (iv) residual non-Markovian memory in the organized cluster—are genuine physical features or artifacts of the chosen cluster count. We refit the analysis on a $K = 5$ model (`exp_20260223_1140_5c129e`; identical VAE backbone, `latent_dim`=8, `hidden_dim`=512) and compare it to the $K = 7$ paper model state by state. Both models are applied to the same 19 475-step time series, so clusters can be mapped directly through their temporal co-occurrence.

2 Cluster structure and 7 → 5 mapping

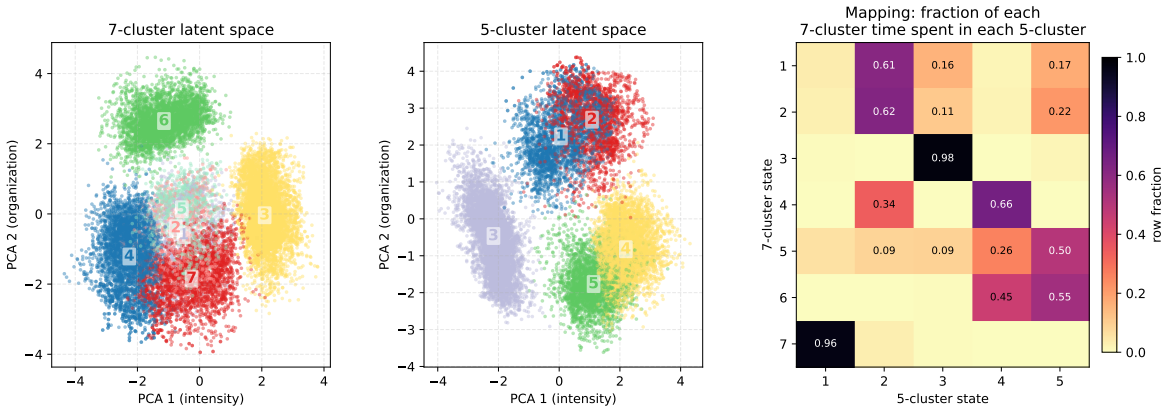


Figure 1: Latent PCA space coloured by cluster for the 7-state (left) and 5-state (middle) models, and the co-occurrence mapping (right): each row is a 7-cluster state, each column a 5-cluster state, and the cell gives the fraction of that 7-cluster's time that the 5-state model assigns to the corresponding 5-cluster (rows sum to 1).

Both models partition the same latent geometry (Fig. 1): a large quiescent core, a moist/organized lobe at high PC2, and a dry, convectively active lobe at low PC2. The mapping is highly structured rather than a reshuffle:

- **Two anchor states are preserved almost 1:1.** The organized, highest-PW cluster (7c 7, PW = 16.9) maps to 5c 1 (PW = 16.9) with fraction 0.96; the large quiescent cluster (7c 3) maps to 5c 3 with fraction 0.98.
- **The moderate-moisture band merges.** 7c 1 and 7c 2 (PW ≈ 9–10, moderate precip) collapse into a single 5c 2 (≈ 0.61–0.62 each).
- **The dry/active band is redistributed.** The heavy-precip 7c 4 maps mostly to 5c 4 (0.66); the high-PC2 active state 7c 6 splits between 5c 5 (0.55) and 5c 4 (0.45); the transitional 7c 5 goes mainly to 5c 5.

So coarsening from 7 to 5 keeps the physically extreme states (organized, quiescent) intact and merges the intermediate moisture/activity band. Table 1 lists the per-cluster fingerprints.

3 Axes of variability: precipitation and organization

In both models PC1 tracks intensity and PC2 tracks organization. The Pearson correlations (signs are arbitrary; magnitudes are comparable) are $|r(\mathrm{PC1}, \mathrm{precip})| = 0.62$ (7c) vs. 0.55 (5c) and $|r(\mathrm{PC2}, \mathrm{PW})| = 0.63$ (7c) vs.

Table 1: Per-cluster fingerprints. n: occupancy; precip and PW: domain diagnostics; π: stationary weight; RMSE: gap between the empirical indicator autocorrelation and the Markov prediction (larger = more non-Markovian).

7-cluster model (PCA EVR = 0.21, 0.18)						
State	n	precip	PW	π	RMSE	note
1	886	3.82	9.35	0.046	0.122	
2	686	4.19	10.26	0.035	0.128	
3	7086	1.68	9.35	0.364	0.078	quiescent
4	3703	5.36	6.07	0.191	0.061	heavy precip
5	777	3.06	5.85	0.040	0.076	
6	3819	3.06	3.55	0.197	**0.254**	organized/active
7	2518	4.74	16.88	0.127	**0.275**	organized (max PW)
5-cluster model (PCA EVR = 0.23, 0.18)						
State	n	precip	PW	π	RMSE	note
1	2562	4.69	16.85	0.129	**0.287**	organized (max PW)
2	2376	4.99	9.85	0.122	0.156	moist, intense precip
3	7230	1.71	9.42	0.370	0.087	quiescent
4	4380	4.52	4.04	0.227	**0.227**	dry/active
5	2927	2.74	4.22	0.152	0.112	dry transitional

0.68 (5c). The precipitation/organization plane is therefore *not* an artifact of cluster count: it is intrinsic to the VAE latent space, which the HMM merely partitions. If anything the organization (PW) axis is slightly better expressed at $K = 5$, and the explained variance of the first two PCs is essentially unchanged (0.39 vs. 0.41).

4 Vertical structure: positive vs. negative mass flux

Decomposing the mass flux into its positive (updraft) and negative (downdraft) parts shows the same physical ordering of clusters in both models (Fig. 2). The organized cluster carries a deep, strong updraft anomaly peaking in the mid-to-upper troposphere (∼8–9 km) together with the largest compensating downdraft anomaly—the signature of deep, vertically coherent convection—while the quiescent cluster is near zero at all levels and the dry/active clusters show shallow, low-level structure. The updraft/downdraft contrast is therefore not an artifact of cluster count.

The one difference mirrors the mapping (§2): at $K = 7$ the deepest, strongest updraft belongs to the organized high-PW state 7c 7, whereas at $K = 5$ that maximal updraft is shared between the organized 5c 1 and the intense-precip 5c 2—the two states that absorb 7c 7 and the moderate-moisture band. The five-state model thus splits the "deep convection" signature across its two moist clusters rather than concentrating it in one.

5 Transition matrices and slow dynamics

Both transition matrices (Fig. 3) are strongly diagonal-dominant and share the same off-diagonal topology: the organized and quiescent states exchange among themselves and with the moist band, while the dry/active states form a separate basin. The relaxation spectra are consistent: the slowest non-trivial mode has $\tau_2 = 144$ min ($K = 7$) vs. 173 min ($K = 5$), and both chains produce a hierarchy of modes from ∼25 min to ∼3 h. Coarsening to 5 states does not collapse or invent slow timescales; it merges fast, weakly-occupied transients (the $K = 7$ chain has two extra modes near $\tau \approx 26$ min).

6 Two-regime leading mode v_2

The two-regime structure survives (Fig. 4). At $K = 5$ the split is sharp: $v_2 < 0$ for the moist/organized group $\{1, 2, 3\}$ and $v_2 > 0$ for the dry group $\{4, 5\}$, i.e. the slowest mode is a moisture/organization see-saw. At $K = 7$ the same contrast dominates—the largest amplitudes are the organized 7c 7 (−0.65) and quiescent 7c 3 (−0.31) against the dry/active 7c 4 (+0.50) and 7c 6 (+0.44)—but the extra weakly-occupied states (7c 1, 7c 2, 7c 5) carry small, mixed-sign amplitudes that blur the partition. The five-state model expresses the same physical mode more cleanly.

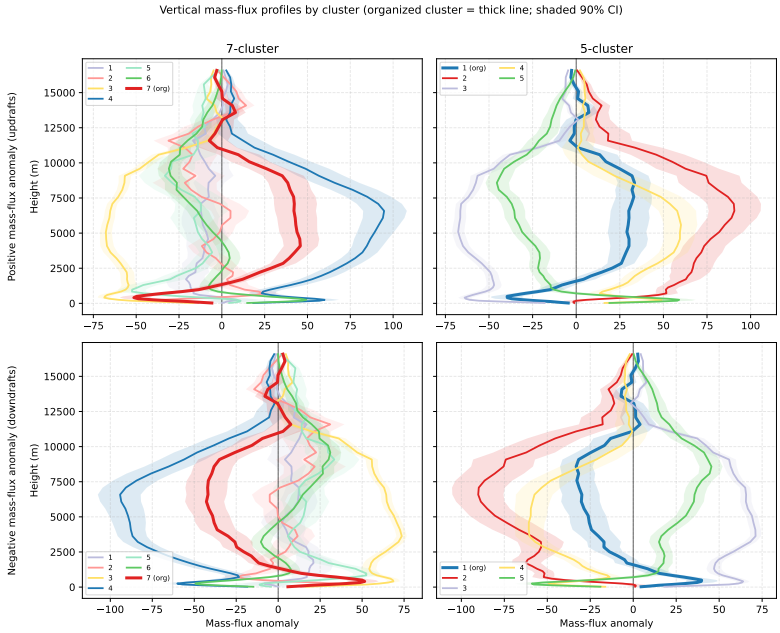


Figure 2: Per-cluster vertical profiles of the positive (top, updrafts) and negative (bottom, downdrafts) mass-flux anomaly, for the 7-state (left) and 5-state (right) models. Profiles are bootstrap means over the closest-to-centroid samples (shaded: 90% CI); the organized (max-PW) cluster is drawn thick.

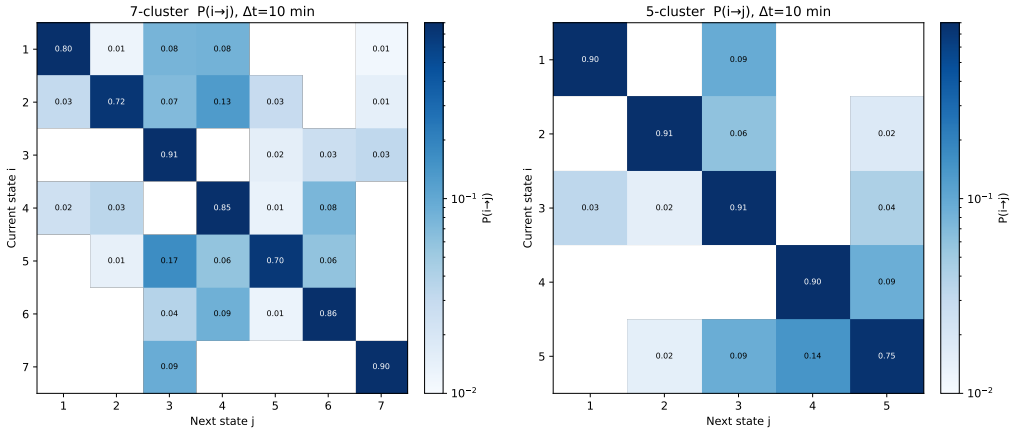


Figure 3: Transition matrices $P(i \to j)$ at the 10-min chain step (log colour scale, entries < 0.01 masked). Both chains are strongly diagonal-dominant (self-persistence 0.75–0.91).

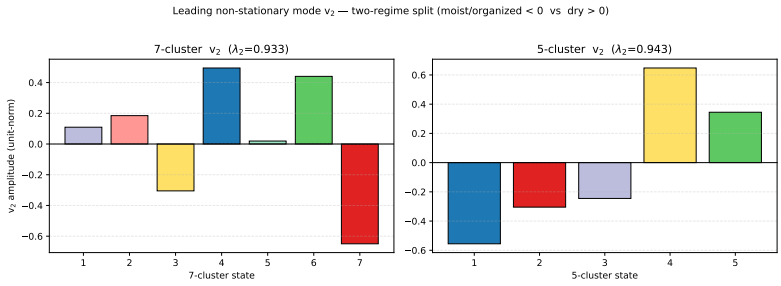


Figure 4: Leading non-stationary right eigenvector v_2 (plain unit-norm amplitudes; the arbitrary sign is fixed so the organized cluster is negative). Both models show a clean two-regime split between the moist/organized side ($v_2 < 0$) and the dry side ($v_2 > 0$).

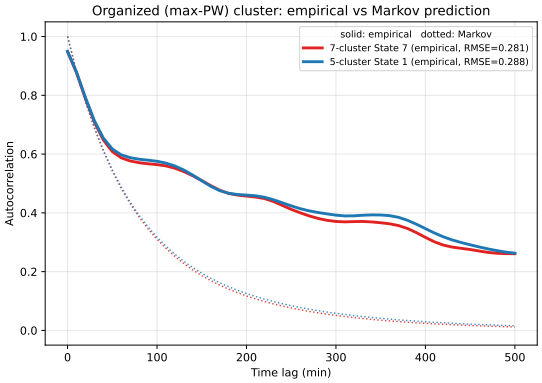


Figure 5: Indicator autocorrelation of the organized (max-PW) cluster: empirical (solid) vs. Markov-chain prediction (dotted), for both models. In both cases the empirical curve persists far above the Markov prediction, signalling residual non-Markovian memory.

7 Markovianity of the organized cluster

The organized cluster is markedly non-Markovian in both models, and to nearly the same degree (Fig. 5): the autocorrelation–Markov RMSE is 0.275 for 7c 7 and 0.287 for 5c 1, with almost identical curves that stay near 0.25 at 500 min while the Markov prediction has decayed to zero. The most Markovian cluster in both models is the large quiescent core (RMSE 0.078/0.087). The one substantive difference is the *second* memory-carrying regime: at $K = 7$ it is a distinct state, 7c 6 (high PC2, RMSE 0.254), whereas at $K = 5$ that memory is absorbed into 5c 4 (RMSE 0.227, fed by 7c 4 and 7c 6 per the mapping). The five-state model retains the strongest non-Markovian signal cleanly but partially merges the two organized memory regimes that $K = 7$ keeps separate.

8 Conclusion: 5 or 7 clusters?

All four headline results are robust to cluster count—the precipitation/organization plane, the organized high-PW cluster, the two-regime v_2 mode, and the non-Markovianity of the organized cluster all appear, with quantitatively similar values, at both $K = 5$ and $K = 7$. None of the paper's qualitative conclusions is an artifact of choosing seven states. **Five clusters is the parsimonious choice**: it preserves the two extreme states almost verbatim (organized 7c 7 ≡ 5c 1, quiescent 7c 3 ≡ 5c 3), gives a slightly cleaner v_2 split and a marginally stronger PC2–PW correlation, and merges only weakly-occupied, fast-relaxing transients. **Seven clusters earns its extra states** in one respect that matters here: it resolves the active/organized lobe into *two* distinct non-Markovian regimes (7c 6 and 7c 7) that $K = 5$ partially merges into 5c 4 and 5c 1.